Trends in Shower Design

by Peter Biermayer

This rain system is sold by Dornbracht, a German company. The promotional brochure that is distributed in the United States includes a disclaimer that says this product does not meet plumbing code.

The latest trend in home improvement seems to be upscale showers. New showering systems range from one straightforward, high-quality showerhead to a custom-built spa that deploys a multitude of showerheads and body sprays. While all of these showering systems are intended to—and often do—provide the consumer with a more pleasant showering experience, marketing for the more deluxe systems suggests that they deliver the type of relaxation previously attributed to whirlpool spas or tubs.

But do they? Is it possible for a shower to deliver bathing nirvana and still meet federal regulations that mandate a maximum flow rate of 2.5 gallons per minute at 80 lb per square inch flowing water pressure? Or are there illegal showerheads and illicit bathers in our newest neighborhoods? The answers are yes, maybe, and it depends on your definition of nirvana.

Some new showerheads use a more sophisticated design to provide consumers with more force, rinsing ability, and coverage, without violating federal regulations. (In shower talk, coverage is the surface area that the spray reaches.) In other showering systems, each showerhead delivers only 2.5 gallons per minute, but two or more showerheads are controlled by a single on-off valve. Then there are blatant violators of the regulations who either advertise their showerheads as using up to 10 gallons per minute, and sometimes even more, or label them as meeting the 2.5-gallon standard, although in actual tests they use much more. Recent tests conducted by the California Energy Commission found that some showerheads labeled 2.0 gallons per minute actually used almost 7 times that much!

Showering System Synopsis

Showers have gotten more sophisticated in recent years. What follows is a review of some of the new showering systems' basic designs and features.

Single-head showerheads. We are all familiar with the standard single-head showerhead. Most deliver no more than 2.5 gallons per minute, and some ultralow-flow models deliver even less. Many come with a choice of settings, including a focused water stream, a wide stream, and a pulsating stream (massage mode). Some are designed to aerate the water and others to deliver parallel streams of water with no air added (laminar flow). Some reduce water flow by using a fixed orifice, while others have a flexible orifice that provides a constant flow rate regardless of the water pressure.

Within this category, the cascade, or downpour, models are gaining in popularity. These models have a large flat head that can be mounted either on an existing shower arm or on the wall, to provide a more vertical flow (see photo, p. 21). They are designed for people who want a gentle, rainlike shower with greater coverage.

Multiple-head showerheads. Multiple-head models have several showerheads connected to a fixture that would normally hold one showerhead. Each individual showerhead may meet federal requirements, but the multiple heads may be connected to a single valve. If these heads cannot be controlled individually, this arrangement skirts the intent of the maximum-flow regulation. Each showerhead may have a separate shutoff valve. Depending on the location of the valves and the configuration of the model, it could be likely that all three showerheads would be on at the same time.

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If each of the heads in a multiple-head showerhead cannot be controlled individually, this arrangement skirts the intent of the maximum-flow regulation.

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Cascade-type showerheads are gaining in popularity.

Shower panels or shower towers. A shower panel or a shower tower typically consists of a single showerhead with body sprays projecting horizontal streams of water toward the bather.

These models often include a hand-held showerhead. Some models of this type are designed for the retrofit market and can be plumbed to an existing shower arm. Others are built in, with several pipes feeding individual shutoff valves, each controlling one or more showerheads.

Body spas. A body spa is a high-end shower found mostly in luxury houses or expensive bathroom remodels. It consists of one or more showerheads, in addition to body sprays at various wall levels. Typically, one valve controls two or more body sprays or showerheads. Some body spa systems recirculate the water through a pump system. This type of body spa may come with an integral electric heater that controls the temperature of the water that gets recirculated and sent out toward the bather.

Water tiles and rain systems. Some high-end, custom showers spray water from perforations in tiles, or from the ceiling itself, to give the effect of rain (see photos, pp. 20 and 22).

HOW MUCH WATER? HOW MUCH ENERGY?

Since many of these showering systems are new, we don't yet know how much water and energy they use. There are still too many unknowns to make even an intelligent guesstimate. How many people will use these new showers? How many of the showerheads that use a lot of water are being sold? Will luxury showers be used infrequently, replacing, perhaps, a weekly soak in a whirlpool tub? Or will they be used every day?

Will the showers that consumers like best, perhaps the ones that let them rinse their hair faster, encourage people to take shorter showers? Or will their greater satisfaction encourage people to take longer showers? Will the showers with multiple shutoff valves be used with all the showerheads on at once? Two at a time? One at a time? Are body spas replacing whirlpool tubs, or will they be used in addition to these tubs? Does the flow rate affect the temperature setting of the water and if so, by how much? How many showerheads are currently not in compliance? Finally, are these new showering systems a passing fad or are they the wave of the future? Only when we have the answers to these questions will we know what effect these systems will have on the use of water and of energy.

WHAT ABOUT THE WATER HEATER?

If a shower uses a lot of water, won't the bather run out of hot water? In order to keep the water warm with high-flow-rate showers, higher-capacity water heaters will be needed. This could mean larger tanks; higher water storage temperatures; multiple tankless water heaters; or new high-efficiency, high-fuel-input, condensing water heaters. The choice of water heater, in turn, will have an effect on piping heat loss, tank heat loss, and the total amount of hot water used. Even the shower drain might have to be made larger to handle the drainage requirements of a high-flow-rate shower.

If luxury showers are replacing whirlpool tubs, do they use more water and energy, or do they use less? It's hard to say. Although whirlpool tubs come in different sizes, they use, on average, 60 gallons per fill. If the incoming cold water is at 60°F and the final mixed water is at 105°F, then 45 gallons of 120°F hot water will be needed to fill one of these tubs. If we assume that the average shower lasts eight minutes, and if a body spa uses 10 gallons per minute, total water use for the shower will be 80 gallons. For this example, total hot water requirements of the shower would exceed those of the whirlpool tub. The unknown factor here is the duration of the shower.

GIVING CUSTOMERS WHAT THEY WANT

Manufacturers are aiming to give customers shower systems that meet their needs and wants. One U.S. manufacturer claims that its new showerhead design uses 36% less water—1.6 rather than 2.5 gallons per minute—while delivering a great shower experience. It achieves this feat by producing larger droplets at a higher velocity, and by using a dense spray pattern. Another manufacturer sells a showerhead that spins each water drop and twirls the entire stream for a complete-coverage shower experience. A researcher in Australia has invented a device that, when fitted into existing showerheads, fills each water droplet with a tiny bubble of air. According to the researcher, this device cuts water use by 30% without sacrificing the sensation of having a full and steady stream of water. Hotel chains have also been attempting to meet customer needs by conducting surveys—asking guests what kind of shower they prefer and how they rate their satisfaction with different showerheads. About two-thirds of the customers surveyed wanted more flow, and roughly two-thirds said they wanted more pressure.

The Canadian Standards Association and the American Society of Mechanical Engineers have established a committee to investigate performance standards for showerheads. Australia already has a procedure for testing performance that includes measuring the temperature of the water that the bather experiences, as well as the coverage. Making information on better-performing low-flow and ultralow-flow showerheads available to the consumer may have two favorable results. First, consumers who use this information to guide their purchasing decisions may be more satisfied consumers. And second, these consumers may use less water, and use less energy to heat water.


KOHLER

This custom showering system uses water tiles.


KOHLER

Not all showerheads are created equal, and consumers may not realize the effect that their choice of showerhead has on their water and energy bills.

Reducing Total Water Consumption

A lot of water can go down the drain while the consumer waits for the hot water to get to the shower, but several devices can reduce, or even eliminate, this waste. One such device is an on-demand hot water recirculation system. The consumer activates this system immediately before taking a shower. When turned on, the system circulates the not-yet-warm-enough water that arrives at the shower back to the water heater until the water is hot enough to be enjoyable. At that point the recirculation system turns itself off and stops pumping the water back to the water heater.

Another device is designed for people who turn on the shower and then walk away and find something else to do while they wait for the water to get hot. Often, by the time they return to the shower, gallons of hot water have gone down the drain. This device senses when the hot water has reached the shower and shuts off the flow. It uses no energy; the device contains paraffin wax, which melts from the heat of the water and mechanically closes the valve. The valve can be opened manually when the user is ready to enter the shower.

Not all showerheads are created equal, and neither are all consumers. While some showerheads give consumers everything they want without wasting energy or water, other showerheads are water and energy hogs. Some consumers are woefully ignorant of the effect that their choice of showerhead has on their water and energy bills. Other consumers are shopping and showering with their eyes wide open. More research is needed to answer the many as yet unanswered questions about showering systems—and perhaps to create standards for ranking showerhead performance. All this will help consumers to buy what they want, and to know exactly what they are buying.

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For more information:

To download a report on showers and water conservation, visit the California Urban Water Conservation Council Web site at www.cuwcc.org/faucets_showerheads.lasso.